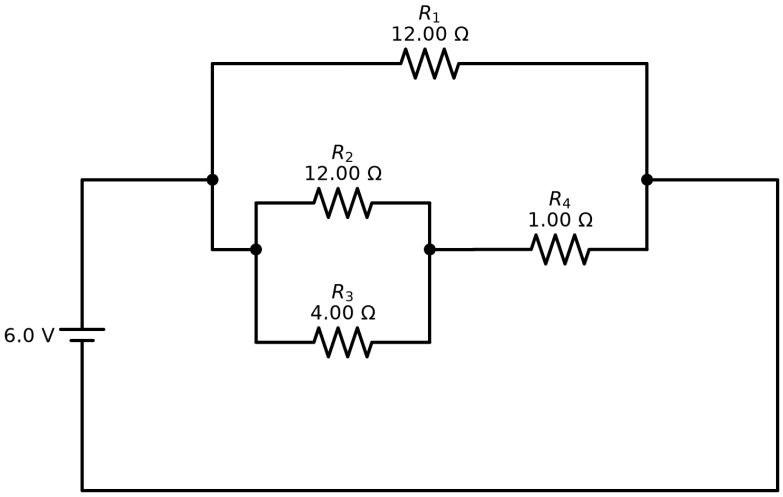
Series-parallel resistor networks: total current (3 reduction steps)

Bank PHY2-DC-SPRTC3-10232026 · created 10-23-2026 · 30 problems

Isomorphic numerical problems. Each item is a different series-parallel topology of a single battery and resistors; the student reduces the network to an equivalent resistance and applies Ohm's law to find the total current. The topology (and hence the reduction sequence) changes between items, not only the numbers.

Problem 1. The circuit shown is powered by a 6 V battery. Determine the total current supplied by the battery of the circuit shown.



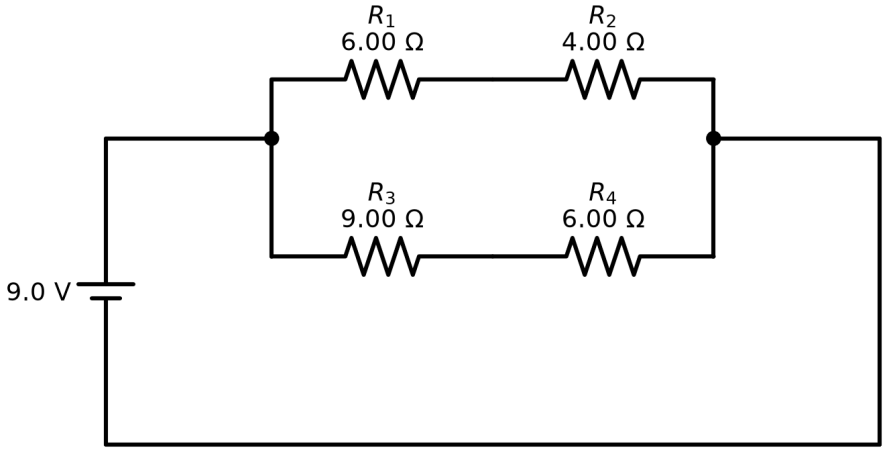
(A) 0.21 A

(B) 7.20 A

(C) 1.50 A

(D) 2.00 A

Problem 2. The circuit shown is powered by a 9 V battery. Determine the total current supplied by the battery of the circuit shown.



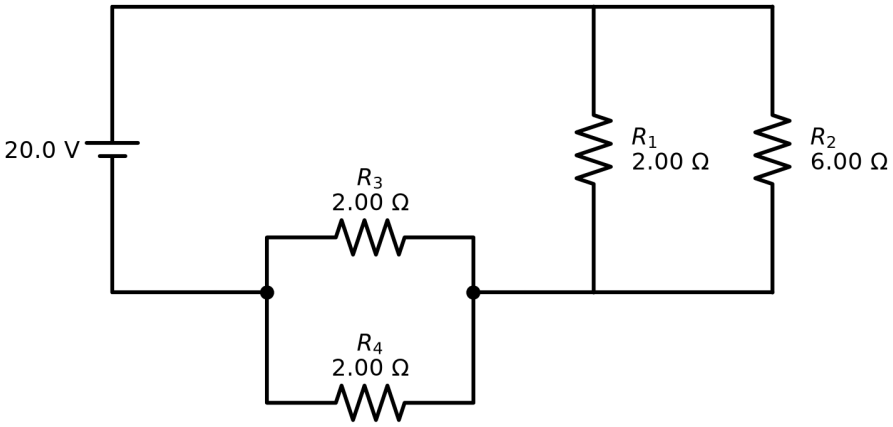
(A) 54.00 A

(B) 0.90 A

(C) 1.50 A

(D) 0.36 A

Problem 3. The circuit shown is powered by a 20 V battery. Determine the total current supplied by the battery of the circuit shown.



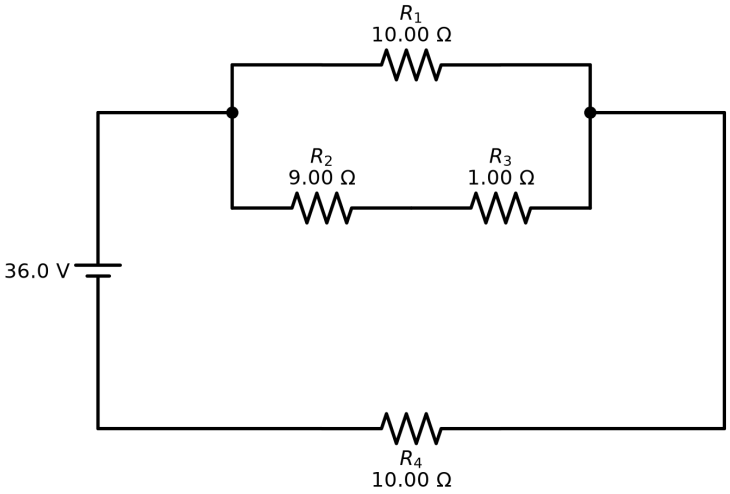
(A) 6.00 A

(B) 8.00 A

(C) 1.67 A

(D) 12.00 A

Problem 4. The circuit shown is powered by a 36 V battery. Determine the total current supplied by the battery of the circuit shown.



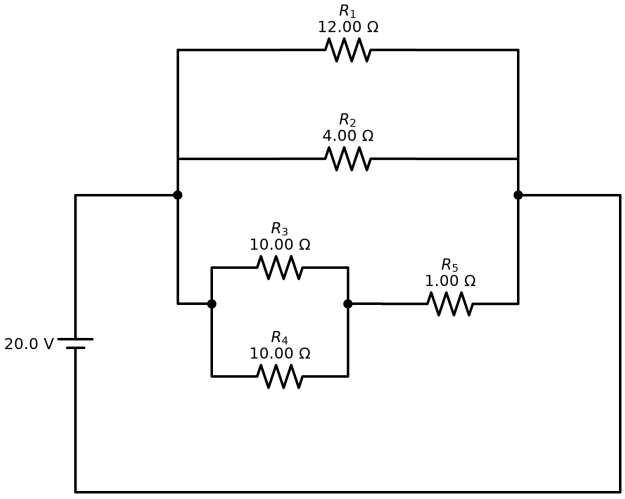
(A) 2.40 A

(B) 1.20 A

(C) 3.53 A

(D) 6.90 A

Problem 5. The circuit shown is powered by a 20 V battery. Determine the total current supplied by the battery of the circuit shown.



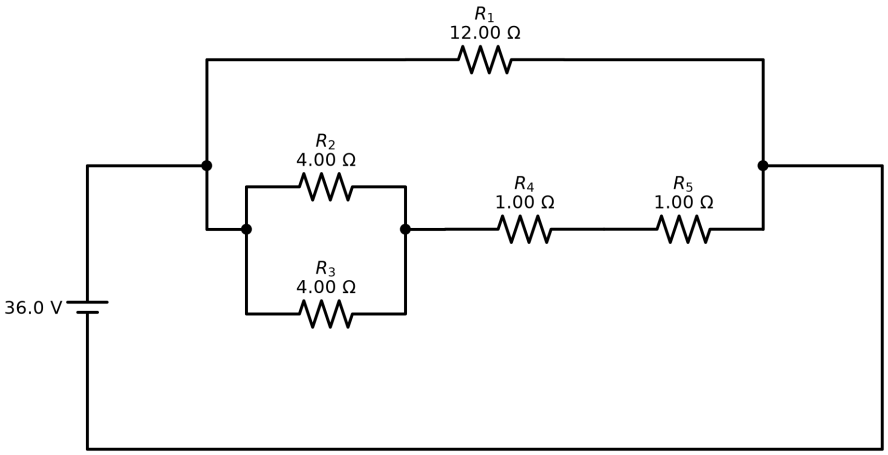
(A) 0.54 A

(B) 17.14 A

(C) 5.00 A

(D) 10.00 A

Problem 6. The circuit shown is powered by a 36 V battery. Determine the total current supplied by the battery of the circuit shown.



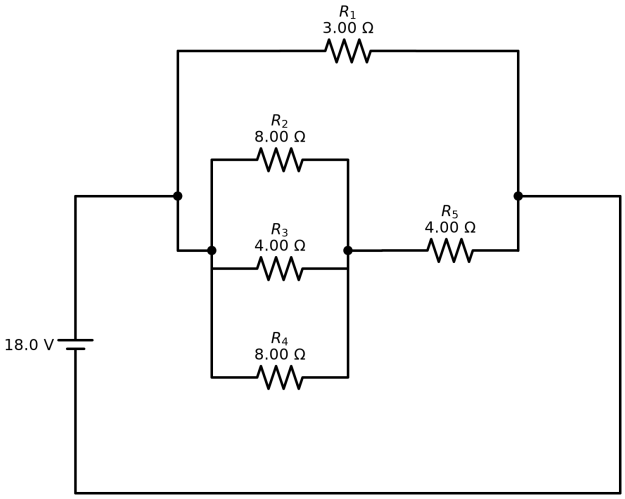
(A) 74.48 A

(B) 9.00 A

(C) 12.00 A

(D) 1.64 A

Problem 7. The circuit shown is powered by a 18 V battery. Determine the total current supplied by the battery of the circuit shown.



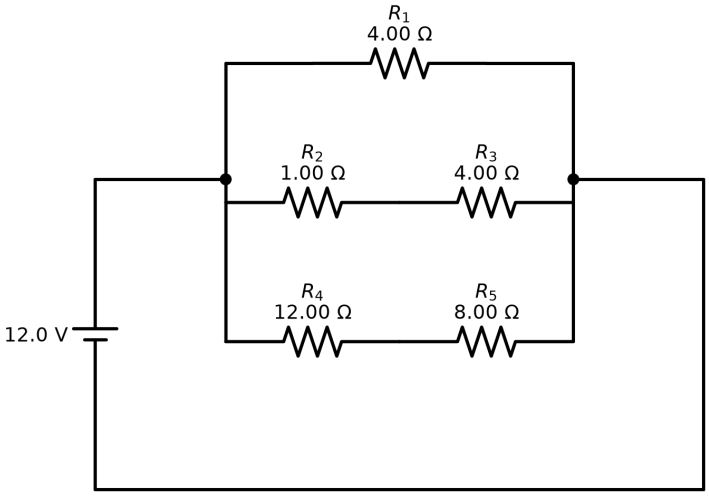
(A) 6.00 A

(B) 9.00 A

(C) 0.67 A

(D) 32.40 A

Problem 8. The circuit shown is powered by a 12 V battery. Determine the total current supplied by the battery of the circuit shown.



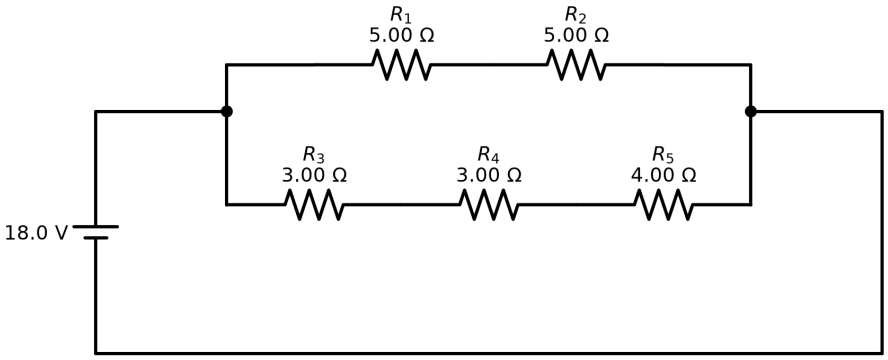
(A) 6.00 A

(B) 0.41 A

(C) 24.00 A

(D) 3.00 A

Problem 9. The circuit shown is powered by a 18 V battery. Determine the total current supplied by the battery of the circuit shown.



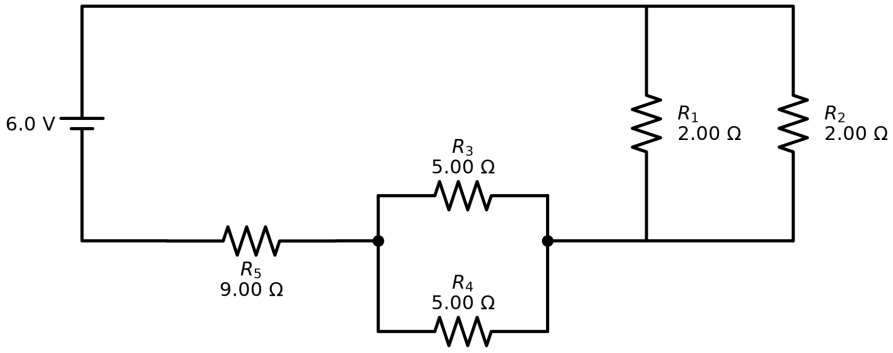
(A) 0.90 A

(B) 90.00 A

(C) 1.80 A

(D) 3.60 A

Problem 10. The circuit shown is powered by a 6 V battery. Determine the total current supplied by the battery of the circuit shown.



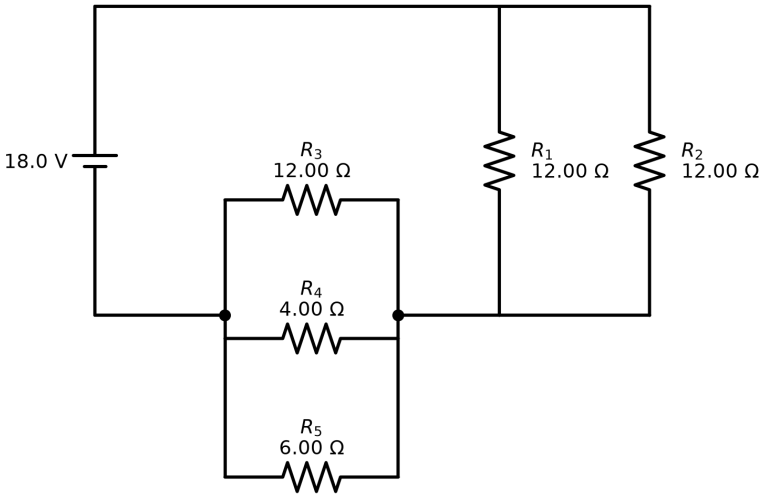
(A) 0.58 A

(B) 0.24 A

(C) 0.48 A

(D) 0.26 A

Problem 11. The circuit shown is powered by a 18 V battery. Determine the total current supplied by the battery of the circuit shown.



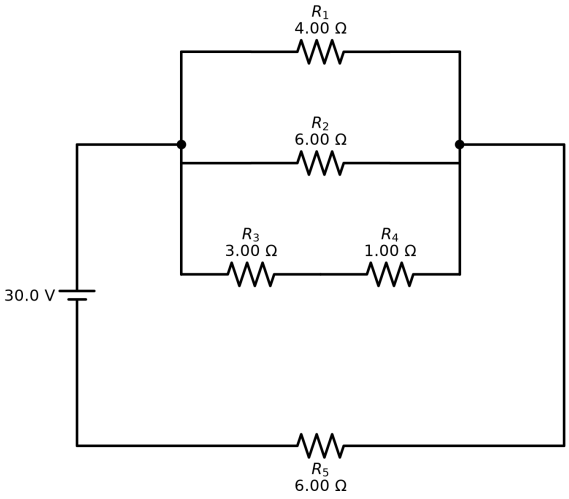
(A) 1.12 A

(B) 2.25 A

(C) 0.39 A

(D) 27.00 A

Problem 12. The circuit shown is powered by a 30 V battery. Determine the total current supplied by the battery of the circuit shown.



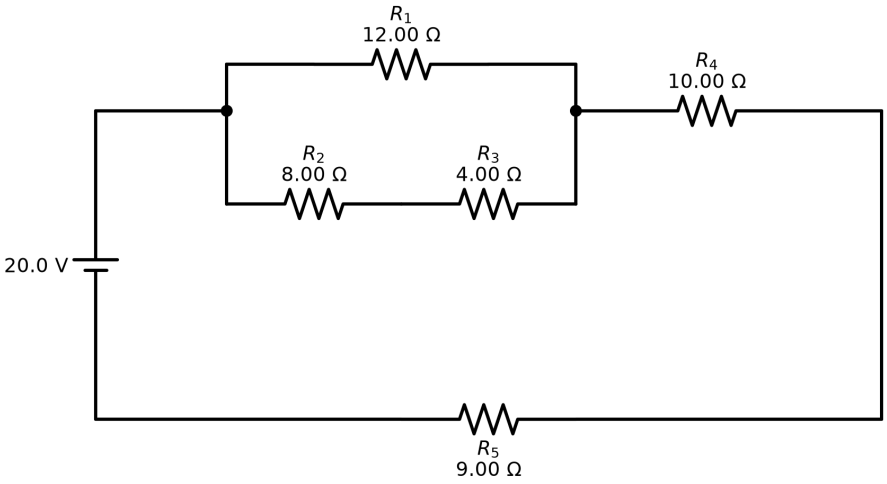
(A) 4.00 A

(B) 1.50 A

(C) 4.50 A

(D) 7.79 A

Problem 13. The circuit shown is powered by a 20 V battery. Determine the total current supplied by the battery of the circuit shown.



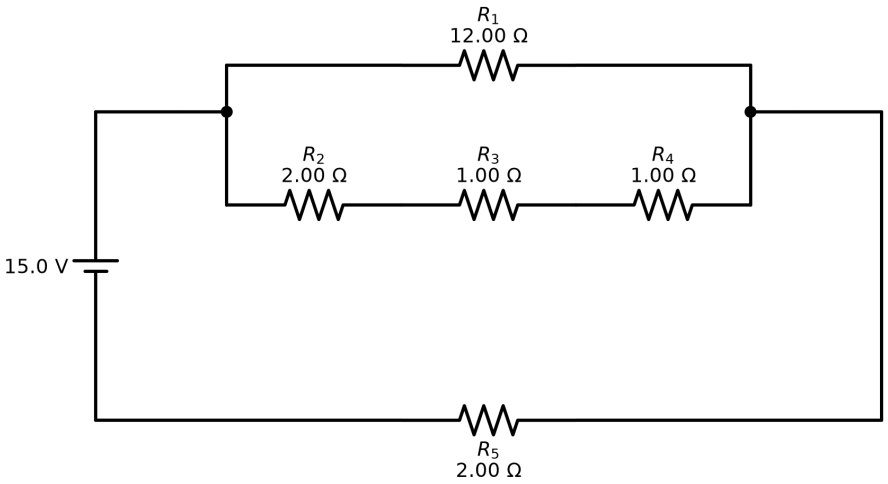
(A) 0.47 A

(B) 1.04 A

(C) 0.40 A

(D) 0.80 A

Problem 14. The circuit shown is powered by a 15 V battery. Determine the total current supplied by the battery of the circuit shown.



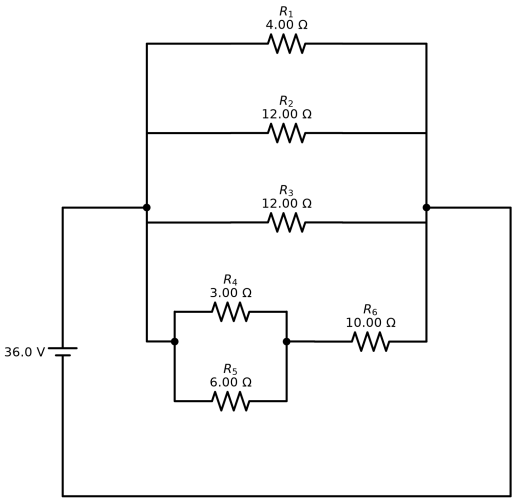
(A) 6.43 A

(B) 2.25 A

(C) 3.00 A

(D) 0.83 A

Problem 15. The circuit shown is powered by a 36 V battery. Determine the total current supplied by the battery of the circuit shown.



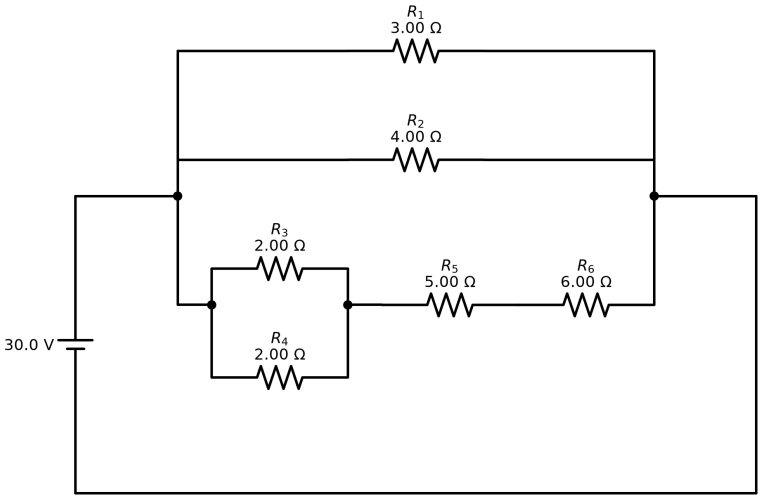
(A) 9.00 A

(B) 18.00 A

(C) 0.77 A

(D) 70.33 A

Problem 16. The circuit shown is powered by a 30 V battery. Determine the total current supplied by the battery of the circuit shown.



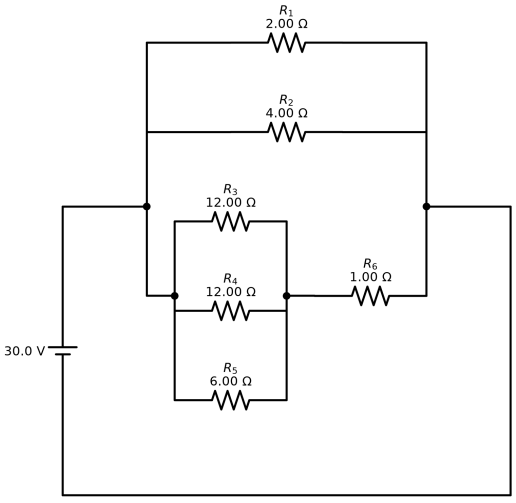
(A) 20.00 A

(B) 1.36 A

(C) 45.00 A

(D) 10.00 A

Problem 17. The circuit shown is powered by a 30 V battery. Determine the total current supplied by the battery of the circuit shown.



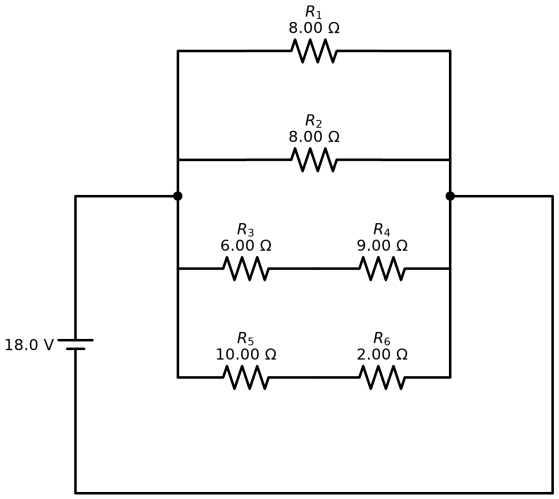
(A) 0.81 A

(B) 20.00 A

(C) 15.00 A

(D) 30.00 A

Problem 18. The circuit shown is powered by a 18 V battery. Determine the total current supplied by the battery of the circuit shown.



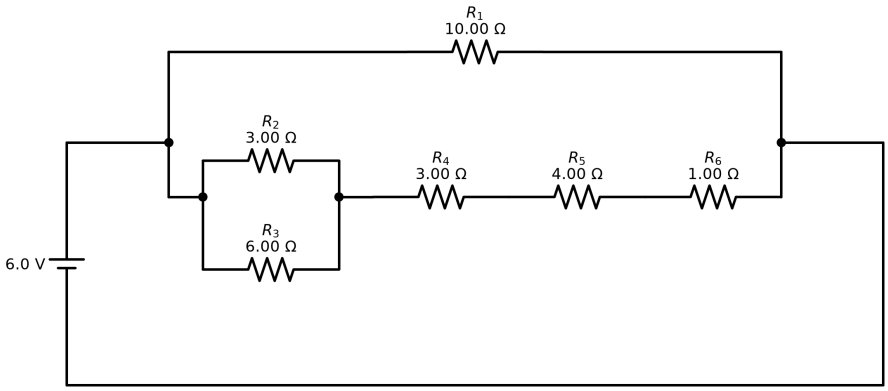
(A) 45.00 A

(B) 2.25 A

(C) 7.20 A

(D) 0.42 A

Problem 19. The circuit shown is powered by a 6 V battery. Determine the total current supplied by the battery of the circuit shown.



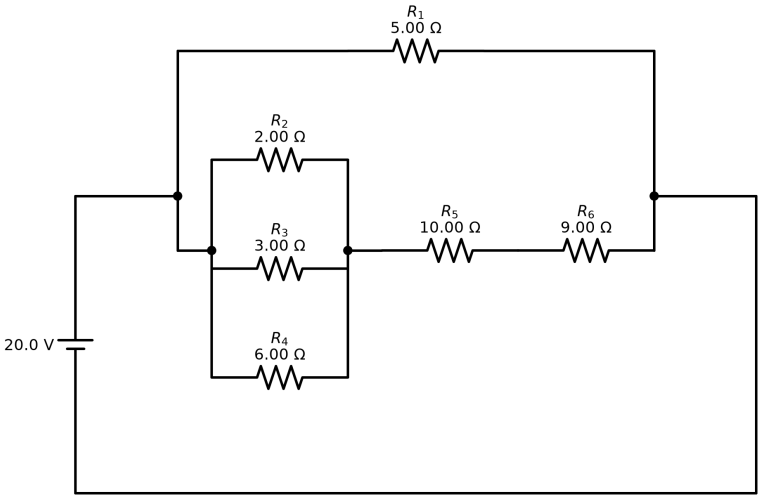
(A) 0.60 A

(B) 1.20 A

(C) 0.22 A

(D) 27.57 A

Problem 20. The circuit shown is powered by a 20 V battery. Determine the total current supplied by the battery of the circuit shown.



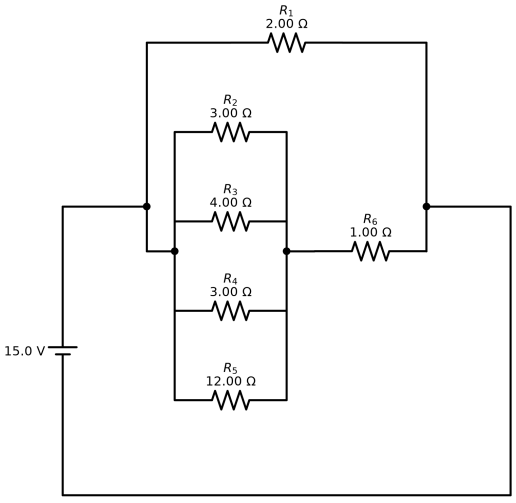
(A) 5.00 A

(B) 0.57 A

(C) 80.00 A

(D) 4.00 A

Problem 21. The circuit shown is powered by a 15 V battery. Determine the total current supplied by the battery of the circuit shown.



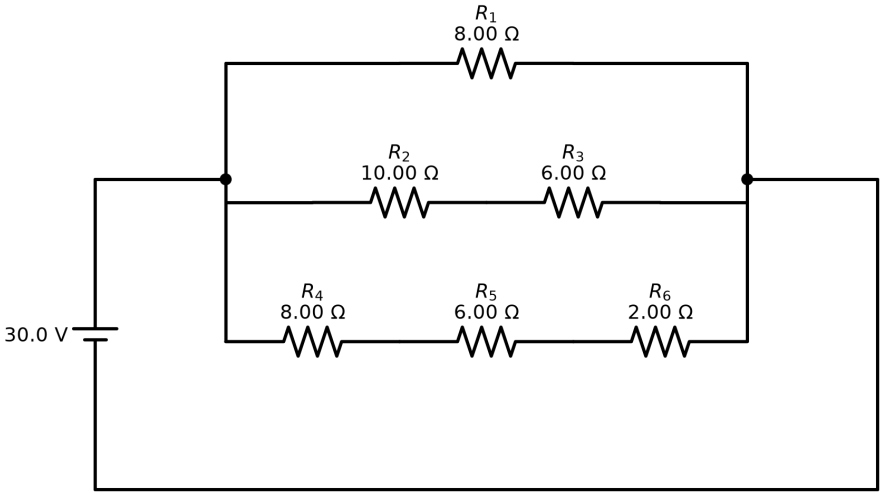
(A) 0.60 A

(B) 7.50 A

(C) 5.07 A

(D) 15.00 A

Problem 22. The circuit shown is powered by a 30 V battery. Determine the total current supplied by the battery of the circuit shown.



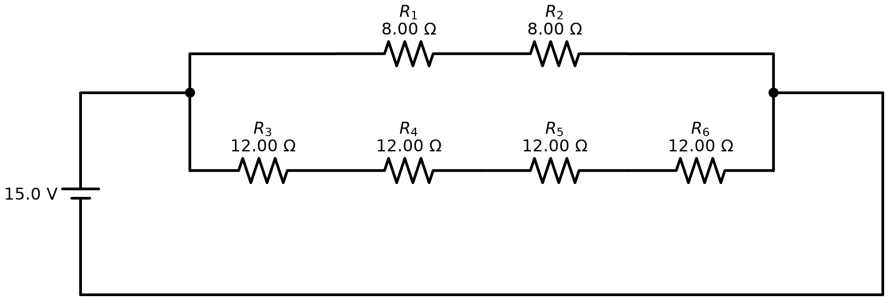
(A) 120.00 A

(B) 3.75 A

(C) 7.50 A

(D) 0.75 A

Problem 23. The circuit shown is powered by a 15 V battery. Determine the total current supplied by the battery of the circuit shown.



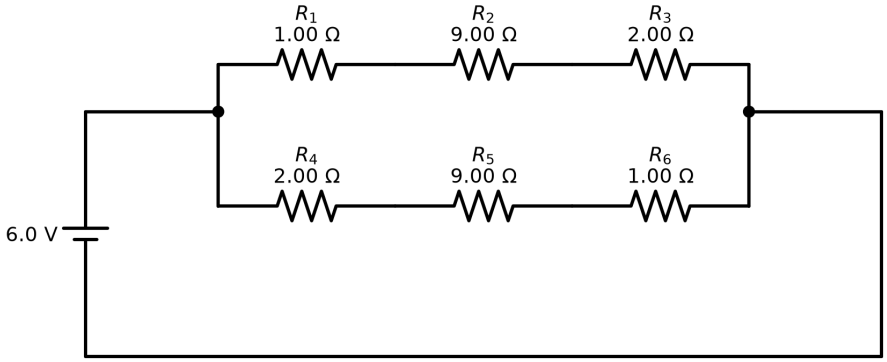
(A) 0.94 A

(B) 1.25 A

(C) 0.23 A

(D) 180.00 A

Problem 24. The circuit shown is powered by a 6 V battery. Determine the total current supplied by the battery of the circuit shown.



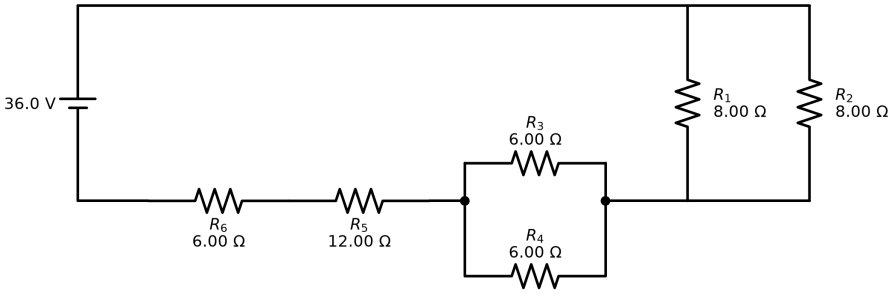
(A) 1.00 A

(B) 0.25 A

(C) 36.00 A

(D) 0.50 A

Problem 25. The circuit shown is powered by a 36 V battery. Determine the total current supplied by the battery of the circuit shown.



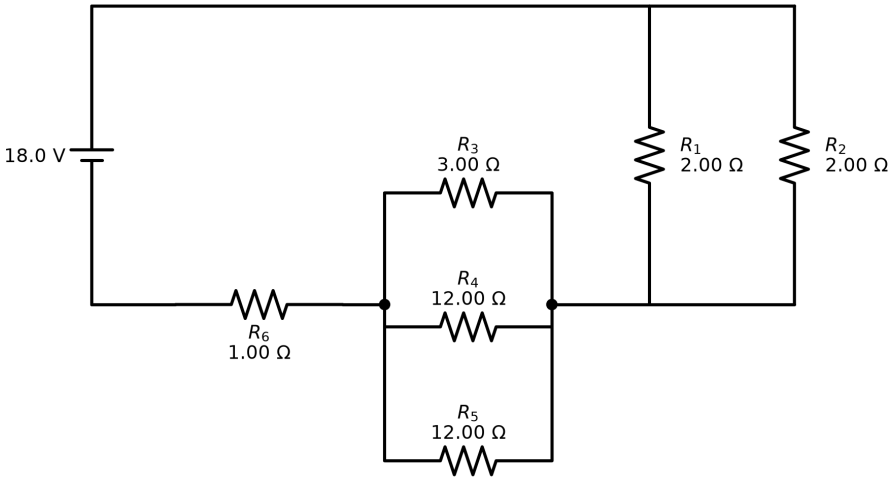
(A) 0.78 A

(B) 1.94 A

(C) 0.72 A

(D) 1.44 A

Problem 26. The circuit shown is powered by a 18 V battery. Determine the total current supplied by the battery of the circuit shown.



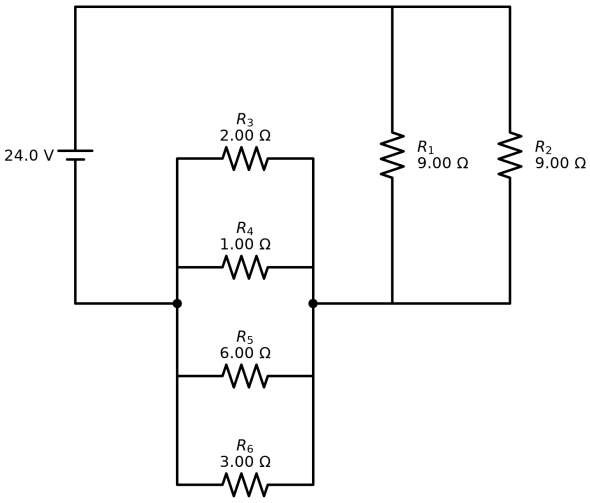
(A) 7.20 A

(B) 3.00 A

(C) 4.50 A

(D) 0.56 A

Problem 27. The circuit shown is powered by a 24 V battery. Determine the total current supplied by the battery of the circuit shown.



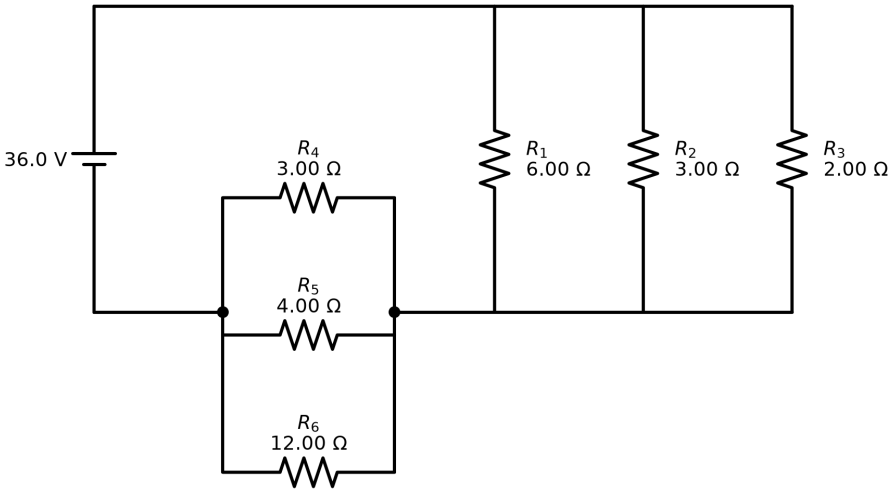
(A) 2.40 A

(B) 4.80 A

(C) 0.80 A

(D) 10.80 A

Problem 28. The circuit shown is powered by a 36 V battery. Determine the total current supplied by the battery of the circuit shown.



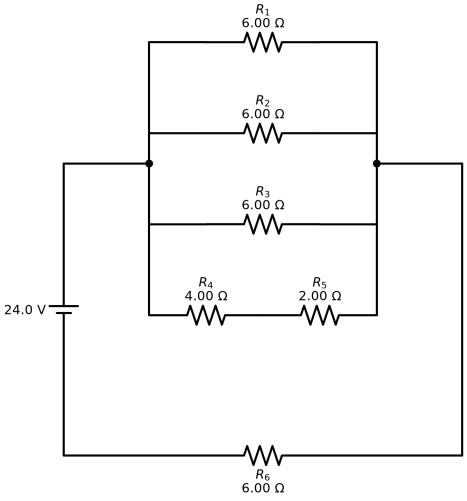
(A) 14.40 A

(B) 1.20 A

(C) 21.60 A

(D) 7.20 A

Problem 29. The circuit shown is powered by a 24 V battery. Determine the total current supplied by the battery of the circuit shown.



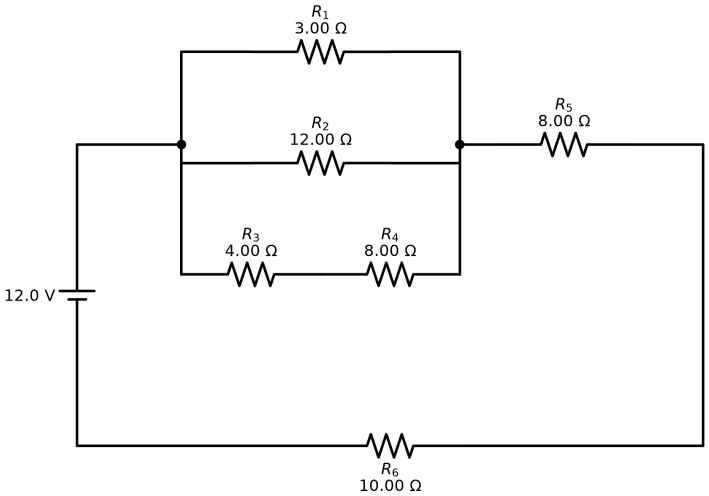
(A) 0.80 A

(B) 3.60 A

(C) 5.24 A

(D) 3.20 A

Problem 30. The circuit shown is powered by a 12 V battery. Determine the total current supplied by the battery of the circuit shown.



(A) 0.65 A

(B) 0.40 A

(C) 0.60 A

(D) 0.27 A

Answer key and worked solutions

Problem 1 — (D) 2.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2 and R3 are in parallel: 1/R23 = 1/12.00 Ω + 1/4.00 Ω, so R23 = 3.00 Ω

2. R23 and R4 are in series: R234 = 3.00 Ω + 1.00 Ω = 4.00 Ω

3. R1 and R234 are in parallel: 1/R1234 = 1/12.00 Ω + 1/4.00 Ω, so R1234 = 3.00 Ω

Work back to the individual parts

1. The battery sees R1234 = 3.00 Ω, so the total current is I = V / R = 6.00 V / 3.00 Ω = 2.00 A.

2. R1 and R234 are in parallel (R1234), so each branch has the same voltage 6.00 V; each current is V / R (0 A through an open branch).

3. R23 and R4 are in series (R234), so each carries the same current 1.50 A; each voltage is I × R (0 V across a wire).

4. R2 and R3 are in parallel (R23), so each branch has the same voltage 4.50 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (12.00 Ω)	0.50 A	6.00 V	3.00 W
R2 (12.00 Ω)	0.38 A	4.50 V	1.69 W
R3 (4.00 Ω)	1.12 A	4.50 V	5.06 W
R4 (1.00 Ω)	1.50 A	1.50 V	2.25 W
Battery	2.00 A	6.00 V	12.00 W (delivered)

Answer

- Total current from the battery = 2.00 A

Answer options (computed, each wrong option is one named mistake)

- 0.21 A: Student adds every resistor as if they were all in series.

- 7.20 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 1.50 A: Student reports the current through R4 (one branch) instead of the total.

- 2.00 A: Correct.

Problem 2 — (C) 1.50 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in series: R12 = 6.00 Ω + 4.00 Ω = 10.00 Ω

2. R3 and R4 are in series: R34 = 9.00 Ω + 6.00 Ω = 15.00 Ω

3. R12 and R34 are in parallel: 1/R1234 = 1/10.00 Ω + 1/15.00 Ω, so R1234 = 6.00 Ω

Work back to the individual parts

1. The battery sees R1234 = 6.00 Ω, so the total current is I = V / R = 9.00 V / 6.00 Ω = 1.50 A.

2. R12 and R34 are in parallel (R1234), so each branch has the same voltage 9.00 V; each current is V / R (0 A through an open branch).

3. R1 and R2 are in series (R12), so each carries the same current 0.90 A; each voltage is I × R (0 V across a wire).

4. R3 and R4 are in series (R34), so each carries the same current 0.60 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (6.00 Ω)	0.90 A	5.40 V	4.86 W
R2 (4.00 Ω)	0.90 A	3.60 V	3.24 W
R3 (9.00 Ω)	0.60 A	5.40 V	3.24 W
R4 (6.00 Ω)	0.60 A	3.60 V	2.16 W
Battery	1.50 A	9.00 V	13.50 W (delivered)

Answer

- Total current from the battery = 1.50 A

Answer options (computed, each wrong option is one named mistake)

- 54.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 0.90 A: Student reports the current through R1 (one branch) instead of the total.

- 1.50 A: Correct.

- 0.36 A: Student adds every resistor as if they were all in series.

Problem 3 — (B) 8.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in parallel: 1/R12 = 1/2.00 Ω + 1/6.00 Ω, so R12 = 1.50 Ω

2. R3 and R4 are in parallel: 1/R34 = 1/2.00 Ω + 1/2.00 Ω, so R34 = 1.00 Ω

3. R12 and R34 are in series: R1234 = 1.50 Ω + 1.00 Ω = 2.50 Ω

Work back to the individual parts

1. The battery sees R1234 = 2.50 Ω, so the total current is I = V / R = 20.00 V / 2.50 Ω = 8.00 A.

2. R12 and R34 are in series (R1234), so each carries the same current 8.00 A; each voltage is I × R (0 V across a wire).

3. R1 and R2 are in parallel (R12), so each branch has the same voltage 12.00 V; each current is V / R (0 A through an open branch).

4. R3 and R4 are in parallel (R34), so each branch has the same voltage 8.00 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (2.00 Ω)	6.00 A	12.00 V	72.00 W
R2 (6.00 Ω)	2.00 A	12.00 V	24.00 W
R3 (2.00 Ω)	4.00 A	8.00 V	32.00 W

R4 (2.00 Ω)	4.00 A	8.00 V	32.00 W
Battery	8.00 A	20.00 V	160.00 W (delivered)

Answer

- Total current from the battery = 8.00 A

Answer options (computed, each wrong option is one named mistake)

- 6.00 A: Student reports the current through R1 (one branch) instead of the total.

- 8.00 A: Correct.

- 1.67 A: Student adds every resistor as if they were all in series.

- 12.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

Problem 4 — (A) 2.40 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2 and R3 are in series: R23 = 9.00 Ω + 1.00 Ω = 10.00 Ω

2. R1 and R23 are in parallel: 1/R123 = 1/10.00 Ω + 1/10.00 Ω, so R123 = 5.00 Ω

3. R123 and R4 are in series: R1234 = 5.00 Ω + 10.00 Ω = 15.00 Ω

Work back to the individual parts

1. The battery sees R1234 = 15.00 Ω, so the total current is I = V / R = 36.00 V / 15.00 Ω = 2.40 A.

2. R123 and R4 are in series (R1234), so each carries the same current 2.40 A; each voltage is I × R (0 V across a wire).

3. R1 and R23 are in parallel (R123), so each branch has the same voltage 12.00 V; each current is V / R (0 A through an open branch).

4. R2 and R3 are in series (R23), so each carries the same current 1.20 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (10.00 Ω)	1.20 A	12.00 V	14.40 W
R2 (9.00 Ω)	1.20 A	10.80 V	12.96 W
R3 (1.00 Ω)	1.20 A	1.20 V	1.44 W
R4 (10.00 Ω)	2.40 A	24.00 V	57.60 W
Battery	2.40 A	36.00 V	86.40 W (delivered)

Answer

- Total current from the battery = 2.40 A

Answer options (computed, each wrong option is one named mistake)

- 2.40 A: Correct.

- 1.20 A: Student adds every resistor as if they were all in series.

- 3.53 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 6.90 A: Student treats series groups as parallel and parallel groups as series.

Problem 5 — (D) 10.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R3 and R4 are in parallel: 1/R34 = 1/10.00 Ω + 1/10.00 Ω, so R34 = 5.00 Ω

2. R34 and R5 are in series: R345 = 5.00 Ω + 1.00 Ω = 6.00 Ω

3. R1, R2 and R345 are in parallel: 1/R12345 = 1/12.00 Ω + 1/4.00 Ω + 1/6.00 Ω, so R12345 = 2.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 2.00 Ω, so the total current is I = V / R = 20.00 V / 2.00 Ω = 10.00 A.

2. R1, R2 and R345 are in parallel (R12345), so each branch has the same voltage 20.00 V; each current is V / R (0 A through an open branch).

3. R34 and R5 are in series (R345), so each carries the same current 3.33 A; each voltage is I × R (0 V across a wire).

4. R3 and R4 are in parallel (R34), so each branch has the same voltage 16.67 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (12.00 Ω)	1.67 A	20.00 V	33.33 W
R2 (4.00 Ω)	5.00 A	20.00 V	100.00 W
R3 (10.00 Ω)	1.67 A	16.67 V	27.78 W
R4 (10.00 Ω)	1.67 A	16.67 V	27.78 W
R5 (1.00 Ω)	3.33 A	3.33 V	11.11 W
Battery	10.00 A	20.00 V	200.00 W (delivered)

Answer

- Total current from the battery = 10.00 A

Answer options (computed, each wrong option is one named mistake)

- 0.54 A: Student adds every resistor as if they were all in series.

- 17.14 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 5.00 A: Student reports the current through R2 (one branch) instead of the total.

- 10.00 A: Correct.

Problem 6 — (C) 12.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2 and R3 are in parallel: 1/R23 = 1/4.00 Ω + 1/4.00 Ω, so R23 = 2.00 Ω

2. R23, R4 and R5 are in series: R2345 = 2.00 Ω + 1.00 Ω + 1.00 Ω = 4.00 Ω

3. R1 and R2345 are in parallel: 1/R12345 = 1/12.00 Ω + 1/4.00 Ω, so R12345 = 3.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 3.00 Ω, so the total current is I = V / R = 36.00 V / 3.00 Ω = 12.00 A.

2. R1 and R2345 are in parallel (R12345), so each branch has the same voltage 36.00 V; each current is V / R (0 A through an open branch).

3. R23, R4 and R5 are in series (R2345), so each carries the same current 9.00 A; each voltage is I × R (0 V across a wire).

4. R2 and R3 are in parallel (R23), so each branch has the same voltage 18.00 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (12.00 Ω)	3.00 A	36.00 V	108.00 W
R2 (4.00 Ω)	4.50 A	18.00 V	81.00 W
R3 (4.00 Ω)	4.50 A	18.00 V	81.00 W
R4 (1.00 Ω)	9.00 A	9.00 V	81.00 W
R5 (1.00 Ω)	9.00 A	9.00 V	81.00 W
Battery	12.00 A	36.00 V	432.00 W (delivered)

Answer

- Total current from the battery = 12.00 A

Answer options (computed, each wrong option is one named mistake)

- 74.48 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 9.00 A: Student reports the current through R4 (one branch) instead of the total.

- 12.00 A: Correct.

- 1.64 A: Student adds every resistor as if they were all in series.

Problem 7 — (B) 9.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2, R3 and R4 are in parallel: 1/R234 = 1/8.00 Ω + 1/4.00 Ω + 1/8.00 Ω, so R234 = 2.00 Ω

2. R234 and R5 are in series: R2345 = 2.00 Ω + 4.00 Ω = 6.00 Ω

3. R1 and R2345 are in parallel: 1/R12345 = 1/3.00 Ω + 1/6.00 Ω, so R12345 = 2.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 2.00 Ω, so the total current is I = V / R = 18.00 V / 2.00 Ω = 9.00 A.

2. R1 and R2345 are in parallel (R12345), so each branch has the same voltage 18.00 V; each current is V / R (0 A through an open branch).

3. R234 and R5 are in series (R2345), so each carries the same current 3.00 A; each voltage is I × R (0 V across a wire).

4. R2, R3 and R4 are in parallel (R234), so each branch has the same voltage 6.00 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (3.00 Ω)	6.00 A	18.00 V	108.00 W
R2 (8.00 Ω)	0.75 A	6.00 V	4.50 W
R3 (4.00 Ω)	1.50 A	6.00 V	9.00 W

R4 (8.00 Ω)	0.75 A	6.00 V	4.50 W
R5 (4.00 Ω)	3.00 A	12.00 V	36.00 W
Battery	9.00 A	18.00 V	162.00 W (delivered)

Answer

- Total current from the battery = 9.00 A

Answer options (computed, each wrong option is one named mistake)

- 6.00 A: Student reports the current through R1 (one branch) instead of the total.

- 9.00 A: Correct.

- 0.67 A: Student adds every resistor as if they were all in series.

- 32.40 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

Problem 8 — (A) 6.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2 and R3 are in series: R23 = 1.00 Ω + 4.00 Ω = 5.00 Ω

2. R4 and R5 are in series: R45 = 12.00 Ω + 8.00 Ω = 20.00 Ω

3. R1, R23 and R45 are in parallel: 1/R12345 = 1/4.00 Ω + 1/5.00 Ω + 1/20.00 Ω, so R12345 = 2.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 2.00 Ω, so the total current is I = V / R = 12.00 V / 2.00 Ω = 6.00 A.

2. R1, R23 and R45 are in parallel (R12345), so each branch has the same voltage 12.00 V; each current is V / R (0 A through an open branch).

3. R2 and R3 are in series (R23), so each carries the same current 2.40 A; each voltage is I × R (0 V across a wire).

4. R4 and R5 are in series (R45), so each carries the same current 0.60 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (4.00 Ω)	3.00 A	12.00 V	36.00 W
R2 (1.00 Ω)	2.40 A	2.40 V	5.76 W
R3 (4.00 Ω)	2.40 A	9.60 V	23.04 W
R4 (12.00 Ω)	0.60 A	7.20 V	4.32 W
R5 (8.00 Ω)	0.60 A	4.80 V	2.88 W
Battery	6.00 A	12.00 V	72.00 W (delivered)

Answer

- Total current from the battery = 6.00 A

Answer options (computed, each wrong option is one named mistake)

- 6.00 A: Correct.

- 0.41 A: Student adds every resistor as if they were all in series.

- 24.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 3.00 A: Student reports the current through R1 (one branch) instead of the total.

Problem 9 — (D) 3.60 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in series: R12 = 5.00 Ω + 5.00 Ω = 10.00 Ω

2. R3, R4 and R5 are in series: R345 = 3.00 Ω + 3.00 Ω + 4.00 Ω = 10.00 Ω

3. R12 and R345 are in parallel: 1/R12345 = 1/10.00 Ω + 1/10.00 Ω, so R12345 = 5.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 5.00 Ω, so the total current is I = V / R = 18.00 V / 5.00 Ω = 3.60 A.

2. R12 and R345 are in parallel (R12345), so each branch has the same voltage 18.00 V; each current is V / R (0 A through an open branch).

3. R1 and R2 are in series (R12), so each carries the same current 1.80 A; each voltage is I × R (0 V across a wire).

4. R3, R4 and R5 are in series (R345), so each carries the same current 1.80 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (5.00 Ω)	1.80 A	9.00 V	16.20 W
R2 (5.00 Ω)	1.80 A	9.00 V	16.20 W
R3 (3.00 Ω)	1.80 A	5.40 V	9.72 W
R4 (3.00 Ω)	1.80 A	5.40 V	9.72 W
R5 (4.00 Ω)	1.80 A	7.20 V	12.96 W
Battery	3.60 A	18.00 V	64.80 W (delivered)

Answer

- Total current from the battery = 3.60 A

Answer options (computed, each wrong option is one named mistake)

- 0.90 A: Student adds every resistor as if they were all in series.

- 90.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 1.80 A: Student reports the current through R1 (one branch) instead of the total.

- 3.60 A: Correct.

Problem 10 — (C) 0.48 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in parallel: 1/R12 = 1/2.00 Ω + 1/2.00 Ω, so R12 = 1.00 Ω

2. R3 and R4 are in parallel: 1/R34 = 1/5.00 Ω + 1/5.00 Ω, so R34 = 2.50 Ω

3. R12, R34 and R5 are in series: R12345 = 1.00 Ω + 2.50 Ω + 9.00 Ω = 12.50 Ω

Work back to the individual parts

1. The battery sees R12345 = 12.50 Ω, so the total current is I = V / R = 6.00 V / 12.50 Ω = 0.48 A.

2. R12, R34 and R5 are in series (R12345), so each carries the same current 0.48 A; each voltage is I × R (0 V across a wire).

3. R1 and R2 are in parallel (R12), so each branch has the same voltage 0.48 V; each current is V / R (0 A through an open branch).

4. R3 and R4 are in parallel (R34), so each branch has the same voltage 1.20 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (2.00 Ω)	0.24 A	0.48 V	0.12 W
R2 (2.00 Ω)	0.24 A	0.48 V	0.12 W
R3 (5.00 Ω)	0.24 A	1.20 V	0.29 W
R4 (5.00 Ω)	0.24 A	1.20 V	0.29 W
R5 (9.00 Ω)	0.48 A	4.32 V	2.07 W
Battery	0.48 A	6.00 V	2.88 W (delivered)

Answer

- Total current from the battery = 0.48 A

Answer options (computed, each wrong option is one named mistake)

- 0.58 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 0.24 A: Student reports the current through R1 (one branch) instead of the total.

- 0.48 A: Correct.

- 0.26 A: Student adds every resistor as if they were all in series.

Problem 11 — (B) 2.25 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in parallel: 1/R12 = 1/12.00 Ω + 1/12.00 Ω, so R12 = 6.00 Ω

2. R3, R4 and R5 are in parallel: 1/R345 = 1/12.00 Ω + 1/4.00 Ω + 1/6.00 Ω, so R345 = 2.00 Ω

3. R12 and R345 are in series: R12345 = 6.00 Ω + 2.00 Ω = 8.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 8.00 Ω, so the total current is I = V / R = 18.00 V / 8.00 Ω = 2.25 A.

2. R12 and R345 are in series (R12345), so each carries the same current 2.25 A; each voltage is I × R (0 V across a wire).

3. R1 and R2 are in parallel (R12), so each branch has the same voltage 13.50 V; each current is V / R (0 A through an open branch).

4. R3, R4 and R5 are in parallel (R345), so each branch has the same voltage 4.50 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (12.00 Ω)	1.12 A	13.50 V	15.19 W

R2 (12.00 Ω)	1.12 A	13.50 V	15.19 W
R3 (12.00 Ω)	0.38 A	4.50 V	1.69 W
R4 (4.00 Ω)	1.12 A	4.50 V	5.06 W
R5 (6.00 Ω)	0.75 A	4.50 V	3.38 W
Battery	2.25 A	18.00 V	40.50 W (delivered)

Answer

- Total current from the battery = 2.25 A

Answer options (computed, each wrong option is one named mistake)

- 1.12 A: Student reports the current through R1 (one branch) instead of the total.

- 2.25 A: Correct.

- 0.39 A: Student adds every resistor as if they were all in series.

- 27.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

Problem 12 — (A) 4.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R3 and R4 are in series: R34 = 3.00 Ω + 1.00 Ω = 4.00 Ω

2. R1, R2 and R34 are in parallel: 1/R1234 = 1/4.00 Ω + 1/6.00 Ω + 1/4.00 Ω, so R1234 = 1.50 Ω

3. R1234 and R5 are in series: R12345 = 1.50 Ω + 6.00 Ω = 7.50 Ω

Work back to the individual parts

1. The battery sees R12345 = 7.50 Ω, so the total current is I = V / R = 30.00 V / 7.50 Ω = 4.00 A.

2. R1234 and R5 are in series (R12345), so each carries the same current 4.00 A; each voltage is I × R (0 V across a wire).

3. R1, R2 and R34 are in parallel (R1234), so each branch has the same voltage 6.00 V; each current is V / R (0 A through an open branch).

4. R3 and R4 are in series (R34), so each carries the same current 1.50 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (4.00 Ω)	1.50 A	6.00 V	9.00 W
R2 (6.00 Ω)	1.00 A	6.00 V	6.00 W
R3 (3.00 Ω)	1.50 A	4.50 V	6.75 W
R4 (1.00 Ω)	1.50 A	1.50 V	2.25 W
R5 (6.00 Ω)	4.00 A	24.00 V	96.00 W
Battery	4.00 A	30.00 V	120.00 W (delivered)

Answer

- Total current from the battery = 4.00 A

Answer options (computed, each wrong option is one named mistake)

- 4.00 A: Correct.

- 1.50 A: Student adds every resistor as if they were all in series.

- 4.50 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 7.79 A: Student treats series groups as parallel and parallel groups as series.

Problem 13 — (D) 0.80 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2 and R3 are in series: R23 = 8.00 Ω + 4.00 Ω = 12.00 Ω

2. R1 and R23 are in parallel: 1/R123 = 1/12.00 Ω + 1/12.00 Ω, so R123 = 6.00 Ω

3. R123, R4 and R5 are in series: R12345 = 6.00 Ω + 10.00 Ω + 9.00 Ω = 25.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 25.00 Ω, so the total current is I = V / R = 20.00 V / 25.00 Ω = 0.80 A.

2. R123, R4 and R5 are in series (R12345), so each carries the same current 0.80 A; each voltage is I × R (0 V across a wire).

3. R1 and R23 are in parallel (R123), so each branch has the same voltage 4.80 V; each current is V / R (0 A through an open branch).

4. R2 and R3 are in series (R23), so each carries the same current 0.40 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (12.00 Ω)	0.40 A	4.80 V	1.92 W
R2 (8.00 Ω)	0.40 A	3.20 V	1.28 W
R3 (4.00 Ω)	0.40 A	1.60 V	0.64 W
R4 (10.00 Ω)	0.80 A	8.00 V	6.40 W
R5 (9.00 Ω)	0.80 A	7.20 V	5.76 W
Battery	0.80 A	20.00 V	16.00 W (delivered)

Answer

- Total current from the battery = 0.80 A

Answer options (computed, each wrong option is one named mistake)

- 0.47 A: Student adds every resistor as if they were all in series.

- 1.04 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 0.40 A: Student reports the current through R1 (one branch) instead of the total.

- 0.80 A: Correct.

Problem 14 — (C) 3.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2, R3 and R4 are in series: R234 = 2.00 Ω + 1.00 Ω + 1.00 Ω = 4.00 Ω

2. R1 and R234 are in parallel: 1/R1234 = 1/12.00 Ω + 1/4.00 Ω, so R1234 = 3.00 Ω

3. R1234 and R5 are in series: R12345 = 3.00 Ω + 2.00 Ω = 5.00 Ω

Work back to the individual parts

1. The battery sees R12345 = 5.00 Ω, so the total current is I = V / R = 15.00 V / 5.00 Ω = 3.00 A.

2. R1234 and R5 are in series (R12345), so each carries the same current 3.00 A; each voltage is I × R (0 V across a wire).

3. R1 and R234 are in parallel (R1234), so each branch has the same voltage 9.00 V; each current is V / R (0 A through an open branch).

4. R2, R3 and R4 are in series (R234), so each carries the same current 2.25 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (12.00 Ω)	0.75 A	9.00 V	6.75 W
R2 (2.00 Ω)	2.25 A	4.50 V	10.12 W
R3 (1.00 Ω)	2.25 A	2.25 V	5.06 W
R4 (1.00 Ω)	2.25 A	2.25 V	5.06 W
R5 (2.00 Ω)	3.00 A	6.00 V	18.00 W
Battery	3.00 A	15.00 V	45.00 W (delivered)

Answer

- Total current from the battery = 3.00 A

Answer options (computed, each wrong option is one named mistake)

- 6.43 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 2.25 A: Student reports the current through R2 (one branch) instead of the total.

- 3.00 A: Correct.

- 0.83 A: Student adds every resistor as if they were all in series.

Problem 15 — (B) 18.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R4 and R5 are in parallel: 1/R45 = 1/3.00 Ω + 1/6.00 Ω, so R45 = 2.00 Ω

2. R45 and R6 are in series: R456 = 2.00 Ω + 10.00 Ω = 12.00 Ω

3. R1, R2, R3 and R456 are in parallel: 1/R123456 = 1/4.00 Ω + 1/12.00 Ω + 1/12.00 Ω + 1/12.00 Ω, so R123456 = 2.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 2.00 Ω, so the total current is I = V / R = 36.00 V / 2.00 Ω = 18.00 A.

2. R1, R2, R3 and R456 are in parallel (R123456), so each branch has the same voltage 36.00 V; each current is V / R (0 A through an open branch).

3. R45 and R6 are in series (R456), so each carries the same current 3.00 A; each voltage is I × R (0 V across a wire).

4. R4 and R5 are in parallel (R45), so each branch has the same voltage 6.00 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (4.00 Ω)	9.00 A	36.00 V	324.00 W
R2 (12.00 Ω)	3.00 A	36.00 V	108.00 W
R3 (12.00 Ω)	3.00 A	36.00 V	108.00 W
R4 (3.00 Ω)	2.00 A	6.00 V	12.00 W
R5 (6.00 Ω)	1.00 A	6.00 V	6.00 W
R6 (10.00 Ω)	3.00 A	30.00 V	90.00 W
Battery	18.00 A	36.00 V	648.00 W (delivered)

Answer

- Total current from the battery = 18.00 A

Answer options (computed, each wrong option is one named mistake)

- 9.00 A: Student reports the current through R1 (one branch) instead of the total.

- 18.00 A: Correct.

- 0.77 A: Student adds every resistor as if they were all in series.

- 70.33 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

Problem 16 — (A) 20.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R3 and R4 are in parallel: 1/R34 = 1/2.00 Ω + 1/2.00 Ω, so R34 = 1.00 Ω

2. R34, R5 and R6 are in series: R3456 = 1.00 Ω + 5.00 Ω + 6.00 Ω = 12.00 Ω

3. R1, R2 and R3456 are in parallel: 1/R123456 = 1/3.00 Ω + 1/4.00 Ω + 1/12.00 Ω, so R123456 = 1.50 Ω

Work back to the individual parts

1. The battery sees R123456 = 1.50 Ω, so the total current is I = V / R = 30.00 V / 1.50 Ω = 20.00 A.

2. R1, R2 and R3456 are in parallel (R123456), so each branch has the same voltage 30.00 V; each current is V / R (0 A through an open branch).

3. R34, R5 and R6 are in series (R3456), so each carries the same current 2.50 A; each voltage is I × R (0 V across a wire).

4. R3 and R4 are in parallel (R34), so each branch has the same voltage 2.50 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (3.00 Ω)	10.00 A	30.00 V	300.00 W
R2 (4.00 Ω)	7.50 A	30.00 V	225.00 W
R3 (2.00 Ω)	1.25 A	2.50 V	3.12 W
R4 (2.00 Ω)	1.25 A	2.50 V	3.12 W
R5 (5.00 Ω)	2.50 A	12.50 V	31.25 W
R6 (6.00 Ω)	2.50 A	15.00 V	37.50 W
Battery	20.00 A	30.00 V	600.00 W (delivered)

Answer

- Total current from the battery = 20.00 A

Answer options (computed, each wrong option is one named mistake)

- 20.00 A: Correct.

- 1.36 A: Student adds every resistor as if they were all in series.

- 45.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 10.00 A: Student reports the current through R1 (one branch) instead of the total.

Problem 17 — (D) 30.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R3, R4 and R5 are in parallel: 1/R345 = 1/12.00 Ω + 1/12.00 Ω + 1/6.00 Ω, so R345 = 3.00 Ω

2. R345 and R6 are in series: R3456 = 3.00 Ω + 1.00 Ω = 4.00 Ω

3. R1, R2 and R3456 are in parallel: 1/R123456 = 1/2.00 Ω + 1/4.00 Ω + 1/4.00 Ω, so R123456 = 1.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 1.00 Ω, so the total current is I = V / R = 30.00 V / 1.00 Ω = 30.00 A.

2. R1, R2 and R3456 are in parallel (R123456), so each branch has the same voltage 30.00 V; each current is V / R (0 A through an open branch).

3. R345 and R6 are in series (R3456), so each carries the same current 7.50 A; each voltage is I × R (0 V across a wire).

4. R3, R4 and R5 are in parallel (R345), so each branch has the same voltage 22.50 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (2.00 Ω)	15.00 A	30.00 V	450.00 W
R2 (4.00 Ω)	7.50 A	30.00 V	225.00 W
R3 (12.00 Ω)	1.88 A	22.50 V	42.19 W
R4 (12.00 Ω)	1.88 A	22.50 V	42.19 W
R5 (6.00 Ω)	3.75 A	22.50 V	84.38 W
R6 (1.00 Ω)	7.50 A	7.50 V	56.25 W
Battery	30.00 A	30.00 V	900.00 W (delivered)

Answer

- Total current from the battery = 30.00 A

Answer options (computed, each wrong option is one named mistake)

- 0.81 A: Student adds every resistor as if they were all in series.

- 20.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 15.00 A: Student reports the current through R1 (one branch) instead of the total.

- 30.00 A: Correct.

Problem 18 — (C) 7.20 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R3 and R4 are in series: R34 = 6.00 Ω + 9.00 Ω = 15.00 Ω

2. R5 and R6 are in series: R56 = 10.00 Ω + 2.00 Ω = 12.00 Ω

3. R1, R2, R34 and R56 are in parallel: 1/R123456 = 1/8.00 Ω + 1/8.00 Ω + 1/15.00 Ω + 1/12.00 Ω, so R123456 = 2.50 Ω

Work back to the individual parts

1. The battery sees R123456 = 2.50 Ω, so the total current is I = V / R = 18.00 V / 2.50 Ω = 7.20 A.

2. R1, R2, R34 and R56 are in parallel (R123456), so each branch has the same voltage 18.00 V; each current is V / R (0 A through an open branch).

3. R3 and R4 are in series (R34), so each carries the same current 1.20 A; each voltage is I × R (0 V across a wire).

4. R5 and R6 are in series (R56), so each carries the same current 1.50 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (8.00 Ω)	2.25 A	18.00 V	40.50 W
R2 (8.00 Ω)	2.25 A	18.00 V	40.50 W
R3 (6.00 Ω)	1.20 A	7.20 V	8.64 W
R4 (9.00 Ω)	1.20 A	10.80 V	12.96 W
R5 (10.00 Ω)	1.50 A	15.00 V	22.50 W
R6 (2.00 Ω)	1.50 A	3.00 V	4.50 W
Battery	7.20 A	18.00 V	129.60 W (delivered)

Answer

- Total current from the battery = 7.20 A

Answer options (computed, each wrong option is one named mistake)

- 45.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 2.25 A: Student reports the current through R1 (one branch) instead of the total.

- 7.20 A: Correct.

- 0.42 A: Student adds every resistor as if they were all in series.

Problem 19 — (B) 1.20 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2 and R3 are in parallel: 1/R23 = 1/3.00 Ω + 1/6.00 Ω, so R23 = 2.00 Ω

2. R23, R4, R5 and R6 are in series: R23456 = 2.00 Ω + 3.00 Ω + 4.00 Ω + 1.00 Ω = 10.00 Ω

3. R1 and R23456 are in parallel: 1/R123456 = 1/10.00 Ω + 1/10.00 Ω, so R123456 = 5.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 5.00 Ω, so the total current is I = V / R = 6.00 V / 5.00 Ω = 1.20 A.

2. R1 and R23456 are in parallel (R123456), so each branch has the same voltage 6.00 V; each current is V / R (0 A through an open branch).

3. R23, R4, R5 and R6 are in series (R23456), so each carries the same current 0.60 A; each voltage is I × R (0 V across a wire).

4. R2 and R3 are in parallel (R23), so each branch has the same voltage 1.20 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (10.00 Ω)	0.60 A	6.00 V	3.60 W
R2 (3.00 Ω)	0.40 A	1.20 V	0.48 W
R3 (6.00 Ω)	0.20 A	1.20 V	0.24 W
R4 (3.00 Ω)	0.60 A	1.80 V	1.08 W
R5 (4.00 Ω)	0.60 A	2.40 V	1.44 W
R6 (1.00 Ω)	0.60 A	0.60 V	0.36 W
Battery	1.20 A	6.00 V	7.20 W (delivered)

Answer

- Total current from the battery = 1.20 A

Answer options (computed, each wrong option is one named mistake)

- 0.60 A: Student reports the current through R1 (one branch) instead of the total.

- 1.20 A: Correct.

- 0.22 A: Student adds every resistor as if they were all in series.

- 27.57 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

Problem 20 — (A) 5.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2, R3 and R4 are in parallel: 1/R234 = 1/2.00 Ω + 1/3.00 Ω + 1/6.00 Ω, so R234 = 1.00 Ω

2. R234, R5 and R6 are in series: R23456 = 1.00 Ω + 10.00 Ω + 9.00 Ω = 20.00 Ω

3. R1 and R23456 are in parallel: 1/R123456 = 1/5.00 Ω + 1/20.00 Ω, so R123456 = 4.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 4.00 Ω, so the total current is I = V / R = 20.00 V / 4.00 Ω = 5.00 A.

2. R1 and R23456 are in parallel (R123456), so each branch has the same voltage 20.00 V; each current is V / R (0 A through an open branch).

3. R234, R5 and R6 are in series (R23456), so each carries the same current 1.00 A; each voltage is I × R (0 V across a wire).

4. R2, R3 and R4 are in parallel (R234), so each branch has the same voltage 1.00 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (5.00 Ω)	4.00 A	20.00 V	80.00 W

R2 (2.00 Ω)	0.50 A	1.00 V	0.50 W
R3 (3.00 Ω)	0.33 A	1.00 V	0.33 W
R4 (6.00 Ω)	0.17 A	1.00 V	0.17 W
R5 (10.00 Ω)	1.00 A	10.00 V	10.00 W
R6 (9.00 Ω)	1.00 A	9.00 V	9.00 W
Battery	5.00 A	20.00 V	100.00 W (delivered)

Answer

- Total current from the battery = 5.00 A

Answer options (computed, each wrong option is one named mistake)

- 5.00 A: Correct.

- 0.57 A: Student adds every resistor as if they were all in series.

- 80.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 4.00 A: Student reports the current through R1 (one branch) instead of the total.

Problem 21 — (D) 15.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2, R3, R4 and R5 are in parallel: 1/R2345 = 1/3.00 Ω + 1/4.00 Ω + 1/3.00 Ω + 1/12.00 Ω, so R2345 = 1.00 Ω

2. R2345 and R6 are in series: R23456 = 1.00 Ω + 1.00 Ω = 2.00 Ω

3. R1 and R23456 are in parallel: 1/R123456 = 1/2.00 Ω + 1/2.00 Ω, so R123456 = 1.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 1.00 Ω, so the total current is I = V / R = 15.00 V / 1.00 Ω = 15.00 A.

2. R1 and R23456 are in parallel (R123456), so each branch has the same voltage 15.00 V; each current is V / R (0 A through an open branch).

3. R2345 and R6 are in series (R23456), so each carries the same current 7.50 A; each voltage is I × R (0 V across a wire).

4. R2, R3, R4 and R5 are in parallel (R2345), so each branch has the same voltage 7.50 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (2.00 Ω)	7.50 A	15.00 V	112.50 W
R2 (3.00 Ω)	2.50 A	7.50 V	18.75 W
R3 (4.00 Ω)	1.88 A	7.50 V	14.06 W
R4 (3.00 Ω)	2.50 A	7.50 V	18.75 W
R5 (12.00 Ω)	0.62 A	7.50 V	4.69 W
R6 (1.00 Ω)	7.50 A	7.50 V	56.25 W
Battery	15.00 A	15.00 V	225.00 W (delivered)

Answer

- Total current from the battery = 15.00 A

Answer options (computed, each wrong option is one named mistake)

- 0.60 A: Student adds every resistor as if they were all in series.

- 7.50 A: Student reports the current through R1 (one branch) instead of the total.

- 5.07 A: Student treats series groups as parallel and parallel groups as series.

- 15.00 A: Correct.

Problem 22 — (C) 7.50 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R2 and R3 are in series: R23 = 10.00 Ω + 6.00 Ω = 16.00 Ω

2. R4, R5 and R6 are in series: R456 = 8.00 Ω + 6.00 Ω + 2.00 Ω = 16.00 Ω

3. R1, R23 and R456 are in parallel: 1/R123456 = 1/8.00 Ω + 1/16.00 Ω + 1/16.00 Ω, so R123456 = 4.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 4.00 Ω, so the total current is I = V / R = 30.00 V / 4.00 Ω = 7.50 A.

2. R1, R23 and R456 are in parallel (R123456), so each branch has the same voltage 30.00 V; each current is V / R (0 A through an open branch).

3. R2 and R3 are in series (R23), so each carries the same current 1.88 A; each voltage is I × R (0 V across a wire).

4. R4, R5 and R6 are in series (R456), so each carries the same current 1.88 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (8.00 Ω)	3.75 A	30.00 V	112.50 W
R2 (10.00 Ω)	1.88 A	18.75 V	35.16 W
R3 (6.00 Ω)	1.88 A	11.25 V	21.09 W
R4 (8.00 Ω)	1.88 A	15.00 V	28.12 W
R5 (6.00 Ω)	1.88 A	11.25 V	21.09 W
R6 (2.00 Ω)	1.88 A	3.75 V	7.03 W
Battery	7.50 A	30.00 V	225.00 W (delivered)

Answer

- Total current from the battery = 7.50 A

Answer options (computed, each wrong option is one named mistake)

- 120.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 3.75 A: Student reports the current through R1 (one branch) instead of the total.

- 7.50 A: Correct.

- 0.75 A: Student adds every resistor as if they were all in series.

Problem 23 — (B) 1.25 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in series: R12 = 8.00 Ω + 8.00 Ω = 16.00 Ω

2. R3, R4, R5 and R6 are in series: R3456 = 12.00 Ω + 12.00 Ω + 12.00 Ω + 12.00 Ω = 48.00 Ω

3. R12 and R3456 are in parallel: 1/R123456 = 1/16.00 Ω + 1/48.00 Ω, so R123456 = 12.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 12.00 Ω, so the total current is I = V / R = 15.00 V / 12.00 Ω = 1.25 A.

2. R12 and R3456 are in parallel (R123456), so each branch has the same voltage 15.00 V; each current is V / R (0 A through an open branch).

3. R1 and R2 are in series (R12), so each carries the same current 0.94 A; each voltage is I × R (0 V across a wire).

4. R3, R4, R5 and R6 are in series (R3456), so each carries the same current 0.31 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (8.00 Ω)	0.94 A	7.50 V	7.03 W
R2 (8.00 Ω)	0.94 A	7.50 V	7.03 W
R3 (12.00 Ω)	0.31 A	3.75 V	1.17 W
R4 (12.00 Ω)	0.31 A	3.75 V	1.17 W
R5 (12.00 Ω)	0.31 A	3.75 V	1.17 W
R6 (12.00 Ω)	0.31 A	3.75 V	1.17 W
Battery	1.25 A	15.00 V	18.75 W (delivered)

Answer

- Total current from the battery = 1.25 A

Answer options (computed, each wrong option is one named mistake)

- 0.94 A: Student reports the current through R1 (one branch) instead of the total.

- 1.25 A: Correct.

- 0.23 A: Student adds every resistor as if they were all in series.

- 180.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

Problem 24 — (A) 1.00 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1, R2 and R3 are in series: R123 = 1.00 Ω + 9.00 Ω + 2.00 Ω = 12.00 Ω

2. R4, R5 and R6 are in series: R456 = 2.00 Ω + 9.00 Ω + 1.00 Ω = 12.00 Ω

3. R123 and R456 are in parallel: 1/R123456 = 1/12.00 Ω + 1/12.00 Ω, so R123456 = 6.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 6.00 Ω, so the total current is I = V / R = 6.00 V / 6.00 Ω = 1.00 A.

2. R123 and R456 are in parallel (R123456), so each branch has the same voltage 6.00 V; each current is V / R (0 A through an open branch).

3. R1, R2 and R3 are in series (R123), so each carries the same current 0.50 A; each voltage is I × R (0 V across a wire).

4. R4, R5 and R6 are in series (R456), so each carries the same current 0.50 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (1.00 Ω)	0.50 A	0.50 V	0.25 W
R2 (9.00 Ω)	0.50 A	4.50 V	2.25 W
R3 (2.00 Ω)	0.50 A	1.00 V	0.50 W
R4 (2.00 Ω)	0.50 A	1.00 V	0.50 W
R5 (9.00 Ω)	0.50 A	4.50 V	2.25 W
R6 (1.00 Ω)	0.50 A	0.50 V	0.25 W
Battery	1.00 A	6.00 V	6.00 W (delivered)

Answer

- Total current from the battery = 1.00 A

Answer options (computed, each wrong option is one named mistake)

- 1.00 A: Correct.

- 0.25 A: Student adds every resistor as if they were all in series.

- 36.00 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 0.50 A: Student reports the current through R1 (one branch) instead of the total.

Problem 25 — (D) 1.44 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in parallel: 1/R12 = 1/8.00 Ω + 1/8.00 Ω, so R12 = 4.00 Ω

2. R3 and R4 are in parallel: 1/R34 = 1/6.00 Ω + 1/6.00 Ω, so R34 = 3.00 Ω

3. R12, R34, R5 and R6 are in series: R123456 = 4.00 Ω + 3.00 Ω + 12.00 Ω + 6.00 Ω = 25.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 25.00 Ω, so the total current is I = V / R = 36.00 V / 25.00 Ω = 1.44 A.

2. R12, R34, R5 and R6 are in series (R123456), so each carries the same current 1.44 A; each voltage is I × R (0 V across a wire).

3. R1 and R2 are in parallel (R12), so each branch has the same voltage 5.76 V; each current is V / R (0 A through an open branch).

4. R3 and R4 are in parallel (R34), so each branch has the same voltage 4.32 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (8.00 Ω)	0.72 A	5.76 V	4.15 W
R2 (8.00 Ω)	0.72 A	5.76 V	4.15 W
R3 (6.00 Ω)	0.72 A	4.32 V	3.11 W
R4 (6.00 Ω)	0.72 A	4.32 V	3.11 W

R5 (12.00 Ω)	1.44 A	17.28 V	24.88 W
R6 (6.00 Ω)	1.44 A	8.64 V	12.44 W
Battery	1.44 A	36.00 V	51.84 W (delivered)

Answer

- Total current from the battery = 1.44 A

Answer options (computed, each wrong option is one named mistake)

- 0.78 A: Student adds every resistor as if they were all in series.

- 1.94 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 0.72 A: Student reports the current through R3 (one branch) instead of the total.

- 1.44 A: Correct.

Problem 26 — (C) 4.50 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in parallel: 1/R12 = 1/2.00 Ω + 1/2.00 Ω, so R12 = 1.00 Ω

2. R3, R4 and R5 are in parallel: 1/R345 = 1/3.00 Ω + 1/12.00 Ω + 1/12.00 Ω, so R345 = 2.00 Ω

3. R12, R345 and R6 are in series: R123456 = 1.00 Ω + 2.00 Ω + 1.00 Ω = 4.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 4.00 Ω, so the total current is I = V / R = 18.00 V / 4.00 Ω = 4.50 A.

2. R12, R345 and R6 are in series (R123456), so each carries the same current 4.50 A; each voltage is I × R (0 V across a wire).

3. R1 and R2 are in parallel (R12), so each branch has the same voltage 4.50 V; each current is V / R (0 A through an open branch).

4. R3, R4 and R5 are in parallel (R345), so each branch has the same voltage 9.00 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (2.00 Ω)	2.25 A	4.50 V	10.12 W
R2 (2.00 Ω)	2.25 A	4.50 V	10.12 W
R3 (3.00 Ω)	3.00 A	9.00 V	27.00 W
R4 (12.00 Ω)	0.75 A	9.00 V	6.75 W
R5 (12.00 Ω)	0.75 A	9.00 V	6.75 W
R6 (1.00 Ω)	4.50 A	4.50 V	20.25 W
Battery	4.50 A	18.00 V	81.00 W (delivered)

Answer

- Total current from the battery = 4.50 A

Answer options (computed, each wrong option is one named mistake)

- 7.20 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 3.00 A: Student reports the current through R3 (one branch) instead of the total.

- 4.50 A: Correct.

- 0.56 A: Student adds every resistor as if they were all in series.

Problem 27 — (B) 4.80 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1 and R2 are in parallel: 1/R12 = 1/9.00 Ω + 1/9.00 Ω, so R12 = 4.50 Ω

2. R3, R4, R5 and R6 are in parallel: 1/R3456 = 1/2.00 Ω + 1/1.00 Ω + 1/6.00 Ω + 1/3.00 Ω, so R3456 = 0.50 Ω

3. R12 and R3456 are in series: R123456 = 4.50 Ω + 0.50 Ω = 5.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 5.00 Ω, so the total current is I = V / R = 24.00 V / 5.00 Ω = 4.80 A.

2. R12 and R3456 are in series (R123456), so each carries the same current 4.80 A; each voltage is I × R (0 V across a wire).

3. R1 and R2 are in parallel (R12), so each branch has the same voltage 21.60 V; each current is V / R (0 A through an open branch).

4. R3, R4, R5 and R6 are in parallel (R3456), so each branch has the same voltage 2.40 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (9.00 Ω)	2.40 A	21.60 V	51.84 W
R2 (9.00 Ω)	2.40 A	21.60 V	51.84 W
R3 (2.00 Ω)	1.20 A	2.40 V	2.88 W
R4 (1.00 Ω)	2.40 A	2.40 V	5.76 W
R5 (6.00 Ω)	0.40 A	2.40 V	0.96 W
R6 (3.00 Ω)	0.80 A	2.40 V	1.92 W
Battery	4.80 A	24.00 V	115.20 W (delivered)

Answer

- Total current from the battery = 4.80 A

Answer options (computed, each wrong option is one named mistake)

- 2.40 A: Student reports the current through R1 (one branch) instead of the total.

- 4.80 A: Correct.

- 0.80 A: Student adds every resistor as if they were all in series.

- 10.80 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

Problem 28 — (A) 14.40 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R1, R2 and R3 are in parallel: 1/R123 = 1/6.00 Ω + 1/3.00 Ω + 1/2.00 Ω, so R123 = 1.00 Ω

2. R4, R5 and R6 are in parallel: 1/R456 = 1/3.00 Ω + 1/4.00 Ω + 1/12.00 Ω, so R456 = 1.50 Ω

3. R123 and R456 are in series: R123456 = 1.00 Ω + 1.50 Ω = 2.50 Ω

Work back to the individual parts

1. The battery sees R123456 = 2.50 Ω, so the total current is I = V / R = 36.00 V / 2.50 Ω = 14.40 A.

2. R123 and R456 are in series (R123456), so each carries the same current 14.40 A; each voltage is I × R (0 V across a wire).

3. R1, R2 and R3 are in parallel (R123), so each branch has the same voltage 14.40 V; each current is V / R (0 A through an open branch).

4. R4, R5 and R6 are in parallel (R456), so each branch has the same voltage 21.60 V; each current is V / R (0 A through an open branch).

Part	Current	Voltage	Power
R1 (6.00 Ω)	2.40 A	14.40 V	34.56 W
R2 (3.00 Ω)	4.80 A	14.40 V	69.12 W
R3 (2.00 Ω)	7.20 A	14.40 V	103.68 W
R4 (3.00 Ω)	7.20 A	21.60 V	155.52 W
R5 (4.00 Ω)	5.40 A	21.60 V	116.64 W
R6 (12.00 Ω)	1.80 A	21.60 V	38.88 W
Battery	14.40 A	36.00 V	518.40 W (delivered)

Answer

- Total current from the battery = 14.40 A

Answer options (computed, each wrong option is one named mistake)

- 14.40 A: Correct.

- 1.20 A: Student adds every resistor as if they were all in series.

- 21.60 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 7.20 A: Student reports the current through R3 (one branch) instead of the total.

Problem 29 — (D) 3.20 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R4 and R5 are in series: R45 = 4.00 Ω + 2.00 Ω = 6.00 Ω

2. R1, R2, R3 and R45 are in parallel: 1/R12345 = 1/6.00 Ω + 1/6.00 Ω + 1/6.00 Ω + 1/6.00 Ω, so R12345 = 1.50 Ω

3. R12345 and R6 are in series: R123456 = 1.50 Ω + 6.00 Ω = 7.50 Ω

Work back to the individual parts

1. The battery sees R123456 = 7.50 Ω, so the total current is I = V / R = 24.00 V / 7.50 Ω = 3.20 A.

2. R12345 and R6 are in series (R123456), so each carries the same current 3.20 A; each voltage is I × R (0 V across a wire).

3. R1, R2, R3 and R45 are in parallel (R12345), so each branch has the same voltage 4.80 V; each current is V / R (0 A through an open branch).

4. R4 and R5 are in series (R45), so each carries the same current 0.80 A; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (6.00 Ω)	0.80 A	4.80 V	3.84 W
R2 (6.00 Ω)	0.80 A	4.80 V	3.84 W
R3 (6.00 Ω)	0.80 A	4.80 V	3.84 W
R4 (4.00 Ω)	0.80 A	3.20 V	2.56 W
R5 (2.00 Ω)	0.80 A	1.60 V	1.28 W
R6 (6.00 Ω)	3.20 A	19.20 V	61.44 W
Battery	3.20 A	24.00 V	76.80 W (delivered)

Answer

- Total current from the battery = 3.20 A

Answer options (computed, each wrong option is one named mistake)

- 0.80 A: Student adds every resistor as if they were all in series.

- 3.60 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 5.24 A: Student treats series groups as parallel and parallel groups as series.

- 3.20 A: Correct.

Problem 30 — (C) 0.60 A

Question. Determine the total current supplied by the battery of the circuit shown.

Reduce the circuit

1. R3 and R4 are in series: R34 = 4.00 Ω + 8.00 Ω = 12.00 Ω

2. R1, R2 and R34 are in parallel: 1/R1234 = 1/3.00 Ω + 1/12.00 Ω + 1/12.00 Ω, so R1234 = 2.00 Ω

3. R1234, R5 and R6 are in series: R123456 = 2.00 Ω + 8.00 Ω + 10.00 Ω = 20.00 Ω

Work back to the individual parts

1. The battery sees R123456 = 20.00 Ω, so the total current is I = V / R = 12.00 V / 20.00 Ω = 0.60 A.

2. R1234, R5 and R6 are in series (R123456), so each carries the same current 0.60 A; each voltage is I × R (0 V across a wire).

3. R1, R2 and R34 are in parallel (R1234), so each branch has the same voltage 1.20 V; each current is V / R (0 A through an open branch).

4. R3 and R4 are in series (R34), so each carries the same current 100.00 mA; each voltage is I × R (0 V across a wire).

Part	Current	Voltage	Power
R1 (3.00 Ω)	0.40 A	1.20 V	0.48 W
R2 (12.00 Ω)	100.00 mA	1.20 V	0.12 W
R3 (4.00 Ω)	100.00 mA	0.40 V	40.00 mW
R4 (8.00 Ω)	100.00 mA	0.80 V	80.00 mW
R5 (8.00 Ω)	0.60 A	4.80 V	2.88 W

R6 (10.00 Ω)	0.60 A	6.00 V	3.60 W
Battery	0.60 A	12.00 V	7.20 W (delivered)

Answer

- Total current from the battery = 0.60 A

Answer options (computed, each wrong option is one named mistake)

- 0.65 A: Student adds reciprocals for the parallel group but forgets to invert the sum.

- 0.40 A: Student reports the current through R1 (one branch) instead of the total.

- 0.60 A: Correct.

- 0.27 A: Student adds every resistor as if they were all in series.